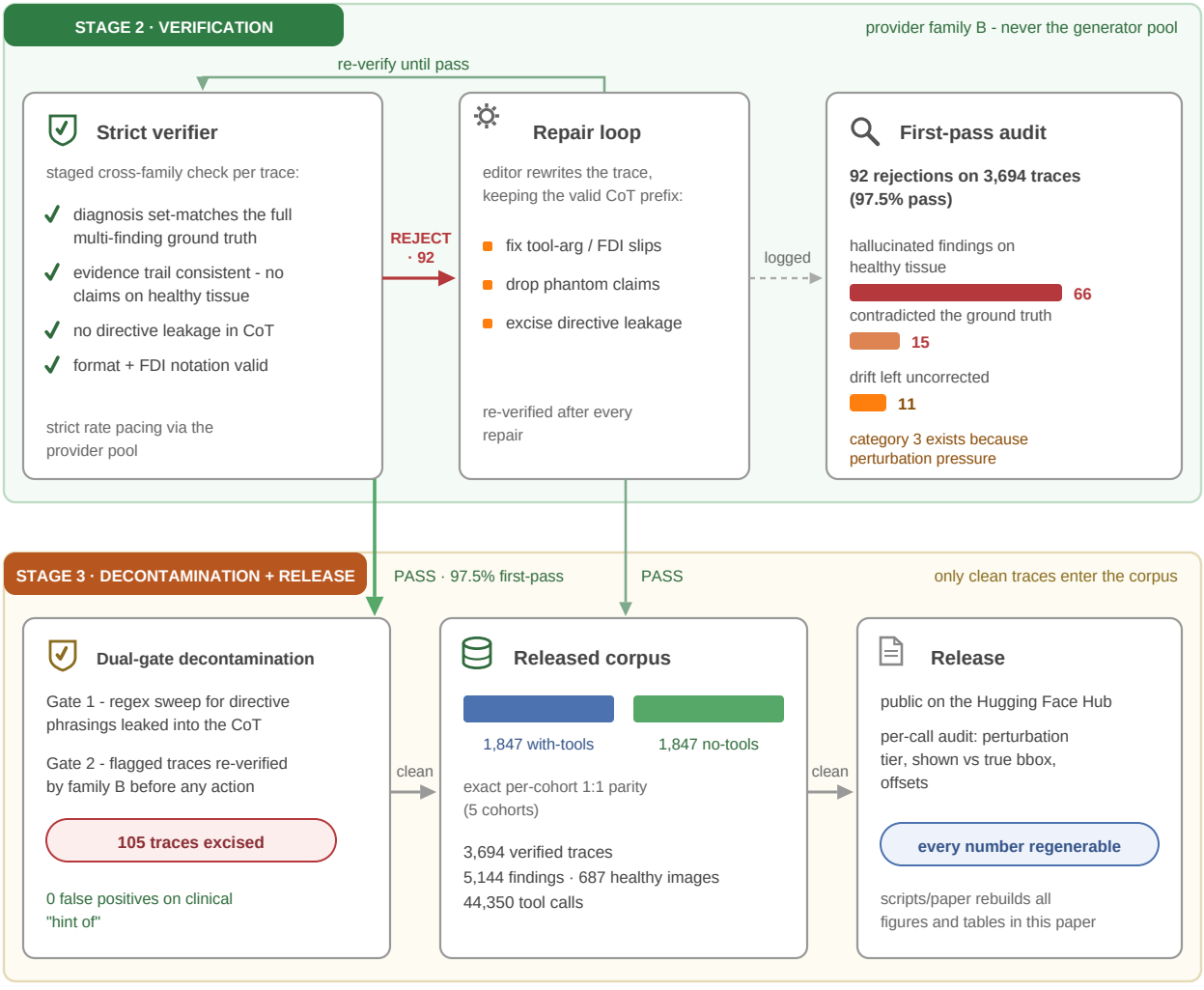
STAGE 2 · VERIFICATION
provider family B - never the generator pool
re-verify until pass
Strict verifier
staged cross-family check per trace:
diagnosis set-matches the full multi-finding ground truth
evidence trail consistent - no claims on healthy tissue
no directive leakage in CoT
format + FDI notation valid
strict rate pacing via the provider pool
REJECT · 92
Repair loop
editor rewrites the trace, keeping the valid CoT prefix:
fix tool-arg / FDI slips
drop phantom claims
excise directive leakage
re-verified after every repair
logged
First-pass audit
92 rejections on 3,694 traces (97.5% pass)
hallucinated findings on healthy tissue
66
contradicted the ground truth
15
drift left uncorrected
11
category 3 exists because perturbation pressure
STAGE 3 · DECONTAMINATION + RELEASE
PASS · 97.5% first-pass
PASS
only clean traces enter the corpus
Dual-gate decontamination
Gate 1 - regex sweep for directive phrasings leaked into the CoT
Gate 2 - flagged traces re-verified by family B before any action
105 traces excised
0 false positives on clinical "hint of"
clean
Released corpus
1,847 with-tools
1,847 no-tools
exact per-cohort 1:1 parity (5 cohorts)
3,694 verified traces
5,144 findings · 687 healthy images
44,350 tool calls
clean
Release
public on the Hugging Face Hub
per-call audit: perturbation tier, shown vs true bbox, offsets
every number regenerable
scripts/paper rebuilds all figures and tables in this paper